HORNSBY GIRLS HIGH SCHOOL

Mathematics Advanced

Year 12 Higher School Certificate
Trial Examination Term 3 2020

STUDENT NUMBER: ______________________

General Instructions:	• Reading time – 10 minutes • Working time – 3 hours • Write using black pen • Calculators approved by NESA may be used • For questions in Section II, show relevant mathematical reasoning and/ or calculations
Total Marks: 100	**Section I – 10 marks** (pages 2–5) • Attempt Questions 1–10 • Allow about 15 minutes for this section **Section II – 90 marks** (pages 6–21) • Attempt Questions 11–34 • Allow about 2 hour and 45 minutes for this section

Section I

10 marks
Attempt questions 1 - 10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1-10

1. What is the value of $f'(x)$ if $f(x) = 3x^4(4-x)^3$?

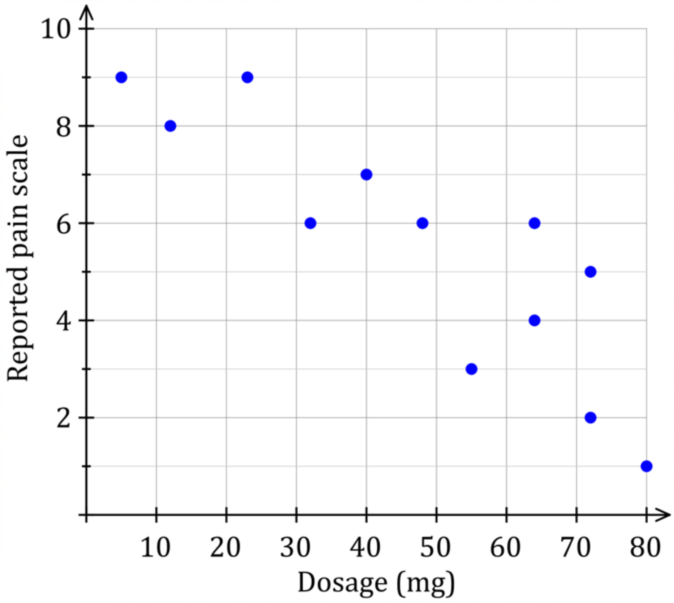
(A) $3x^3(4-x)^3(7x-16)$

(B) $3x^3(4-x)^3(16-7x)$

(C) $3x^3(4-x)^2(7x-16)$

(D) $3x^3(4-x)^2(16-7x)$

2. A scatterplot of pain (as reported by patients) compared to the dosage (in mg) of a drug is shown below.



How could you describe the correlation between the pain and the dosage?

(A) A moderate negative correlation

(B) A moderate positive correlation

(C) A weak positive correlation.

(D) No correlation.

3. What values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

(A) $x < -\frac{1}{6}$

(B) $x > -\frac{1}{6}$

(C) $x < -6$

(D) $x > 6$

4. The probability density function for the continuous random variable X is:

$$f(x) = \begin{cases} x^3 - x + 4 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

The value of $E(X)$ is:

(A) $\dfrac{17}{4}$

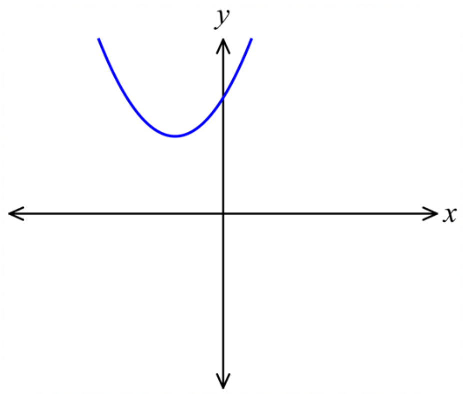
(B) $\dfrac{28}{15}$

(C) 0

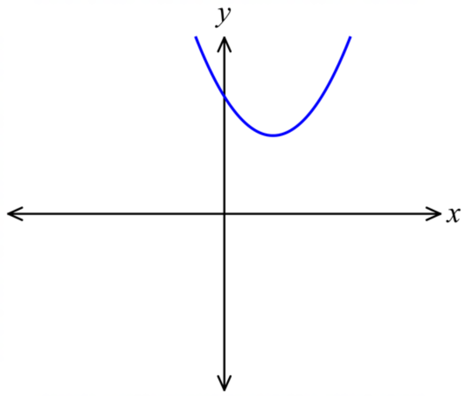
(D) 4

5. Which diagram best shows the graph of the parabola $y = 2 - (x+1)^2$

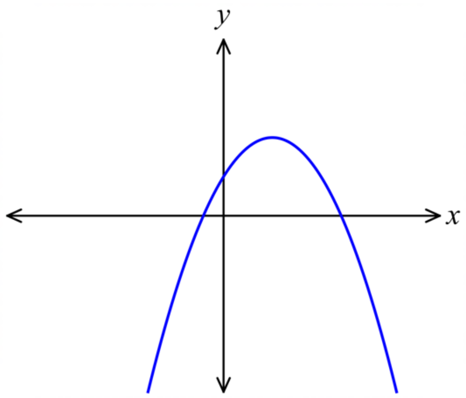
(A)



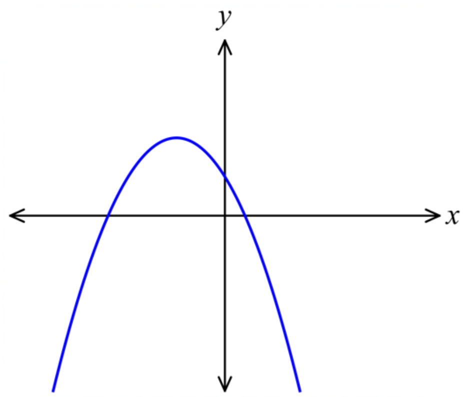
(B)



(C)



(D)



6. Which of the following is the set of all solutions to $2x^2 - 5x + 2 \geq 0$?

(A) $\left[\frac{1}{2}, 2\right]$

(B) $\left(\frac{1}{2}, 2\right)$

(C) $\left(-\infty, \frac{1}{2}\right) \cup (2, \infty)$

(D) $\left(-\infty, \frac{1}{2}\right] \cup [2, \infty)$

7. Which statement is true for an ungrouped data set with no outliers?

(A) The largest possible range is 2 times the interquartile range.

(B) The largest possible range is 3 times the interquartile range.

(C) The largest possible range is 4 times the interquartile range.

(D) The largest possible range is 5 times the interquartile range.

8. A and B are events of a sample space.

   Given that P(A | B) = $4p$, P(B) = p^3 and P(A) = $2p^2$, P(B | A) is equal to

   (A) $2p^2$

   (B) $2p^6$

   (C) $\dfrac{p^2}{2}$

   (D) $\dfrac{2}{p^2}$

9. The continuous random variable X has a normal distribution with a mean 11 and standard deviation 2. If the random variable Z has the standard normal distribution, then the probability that X is greater than 16 is equal to:

   (A) $P(Z > 2)$

   (B) $P(Z < -2.5)$

   (C) $P(Z < 2.5)$

   (D) $P(Z > 5)$

10. What are the solutions to the equation $2\sin x + \sqrt{3} = 0$, where $\{x: 0 \leq x \leq 2\pi\}$?

   (A) $\dfrac{\pi}{3}, \dfrac{2\pi}{3}$

   (B) $\dfrac{2\pi}{3}, \dfrac{5\pi}{3}$

   (C) $\dfrac{4\pi}{3}, \dfrac{5\pi}{3}$

   (D) $\dfrac{7\pi}{3}, \dfrac{11\pi}{3}$

Section II

90 marks
Attempt all questions
Allow about 2 hours and 45 minutes for this section

Answer each question in the spaces provided.
Your responses should include relevant mathematical reasoning and/or calculations.
Extra writing space is provided at the back of the examination paper.

Question 11 (2 marks) **Marks**

Find the primitive function of $\dfrac{1}{1-2x}$ with respect to x. **2**

Question 12 (3 marks)

Let $f(x) = \dfrac{(x+3)(2x+1)}{\sqrt{x}}, x > 0$

(a) By expressing $f(x)$ in the form **2**

$$Ax^{\frac{3}{2}} + Bx^{\frac{1}{2}} + Cx^{-\frac{1}{2}}$$

find the values of A, B and C.

(b) Find $f'(x)$ **1**

Question 13 (3 marks) **Marks**

The discrete random variable, X, has the following probability distribution.

X	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

(a) Find $P(1 < X \leq 3)$ **1**

(b) Find the variance of X. **2**

Question 14 (2 marks)

Find $\int (6x^2 + 2 + x^{-\frac{1}{2}})\,dx$, giving each term in its simplest form. **2**

Question 15 (2 marks)

In an arithmetic sequence, the fifth term is 9 and the twenty-first term is 425. **2**
Find the fiftieth term.

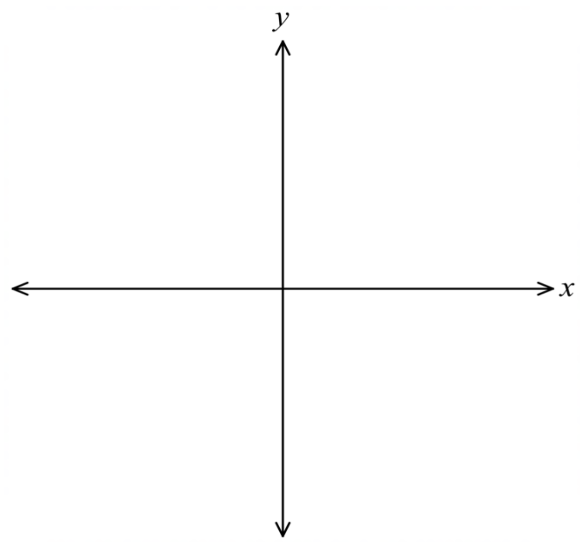
Question 16 (8 marks) **Marks**

Let $f(x) = (x-2)(x^2+1)$

(a) Find where the graph of $y = f(x)$ cuts the x-axis and y axis. **2**

(b) Find the coordinates of the stationary points on the curve with the equation $y = f(x)$ and determine their nature. **3**

(c) Sketch the graphs of $y = f(x)$ and $y = -f(x)$ on the same diagram. **3**



Question 17 (2 marks)

Find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, dx$. **2**

Question 18 (5 marks) **Marks**

Given that $f(x) = (x^2 - 6x)(x - 3) + 2x$.

(a) Express $f(x)$ in the form $x(ax^2 + bx + c)$, where a, b and c are constants. **2**

(b) Hence factorise $f(x)$ completely. **1**

(c) Sketch the graph of $y = f(x)$, showing the coordinates of each point at which the graph meets the axes. **2**

Question 19 (3 marks)

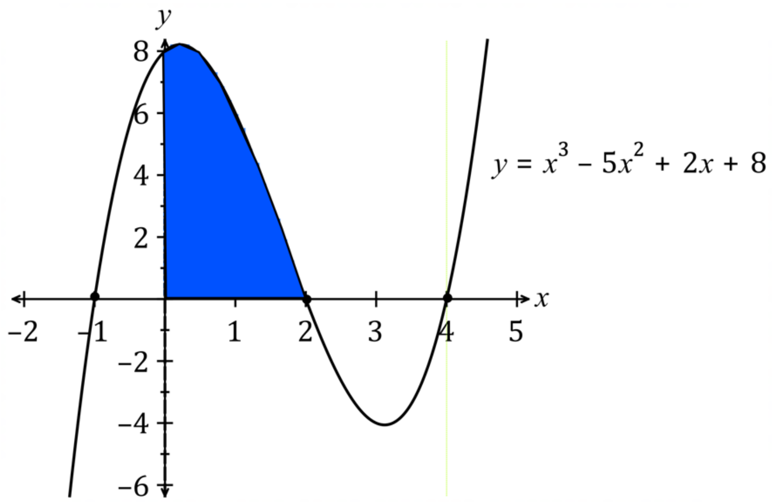
Differentiate with respect to x

(a) $\ln(x^2 + 2)$ **1**

(b) $\dfrac{\sin x}{x^2}$ **2**

Question 20 (2 marks) **Marks**

2



Find the shaded area enclosed by the curve $y = x^3 - 5x^2 + 2x + 8$ and the coordinate axes.

Question 21 (3 marks)

The probability density function for the continuous random variable X is given by:

$$f(x) = \begin{cases} |1 - x| & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find $P(X \leq 1.5)$ **2**

(b) Find $P(1 \leq X \leq 2)$ **1**

Question 22 (5 marks) **Marks**

An arithmetic series is defined as 7+5+3+……………

(a) What is the common difference, d, of this series? **1**

(b) The sum of n terms of this series is -48. **2**

Form an equation in terms of n, and show that your equation may be written as $n^2 - 8n - 48 = 0$.

(c) Find the solution(s) to the equation in part (b). **2**

Question 23 (2 marks)

If $\tan\theta = \dfrac{2}{3}$, and θ is acute, find the exact value of $\sin\theta$? **2**

Question 24 (6 marks) **Marks**

A curve with the equation $y = f(x)$, has $\dfrac{dy}{dx} = x^3 + 2x - 7$.

(a) Find $\dfrac{d^2y}{dx^2}$ **1**

(b) Show that $\dfrac{d^2y}{dx^2} \geq 2$ for all values of x. **1**

(c) The point $P\ (2, 4)$ lies on the curve. Find y in terms of x. **2**

(d) Find an equation for the normal to the curve at P, expressing your answer in the form $ax + by + c = 0$, where a, b and c are integers. **2**

Question 25 (4 marks) **Marks**

The students in a school were surveyed on the number of hours of sleep per week. The results were normally distributed. The survey indicated that 95% of students had between 42 and 54 hours of sleep per week.

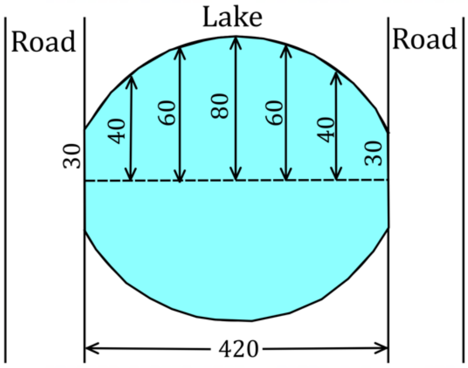
(a) Determine the mean number of hours of sleep per week. **1**

(b) What was the standard deviation? **1**

(c) What percentage of students would have indicated they had between 51 and 57 hours of sleep per week? **2**

Question 26 (3 marks)

A symmetrical lake has two roads, 420 metres apart, forming two of its sides.



Not to scale

Equally spaced measurements of the lake, in metres, are shown on the above diagram. Use the trapezoidal rule to estimate the area of the lake. **3**

Question 27 (7 marks) **Marks**

A particle moves in a straight line. At time t seconds, its distance x metres from a fixed point 0 on the line is given by $x = 1 - \cos 2t$.

(a) Sketch the graph of x as a function of t for $0 \leq t \leq \pi$. **2**

(b) Using your graph, or otherwise, to find the times when the particle is at rest and the position of the particle at these times. **2**

(c) Find the velocity of the particle when $t = \frac{\pi}{4}$. **1**

(d) When is the particle's velocity greater than 1 m/s? **2**

Question 28 (3 marks)

Show that the curve $y = 2x^2 - \ln\left(\frac{x}{2}\right) - 4$ has a minimum value at $\left(\frac{1}{2}, \ln 4 - 3\frac{1}{2}\right)$ **3**

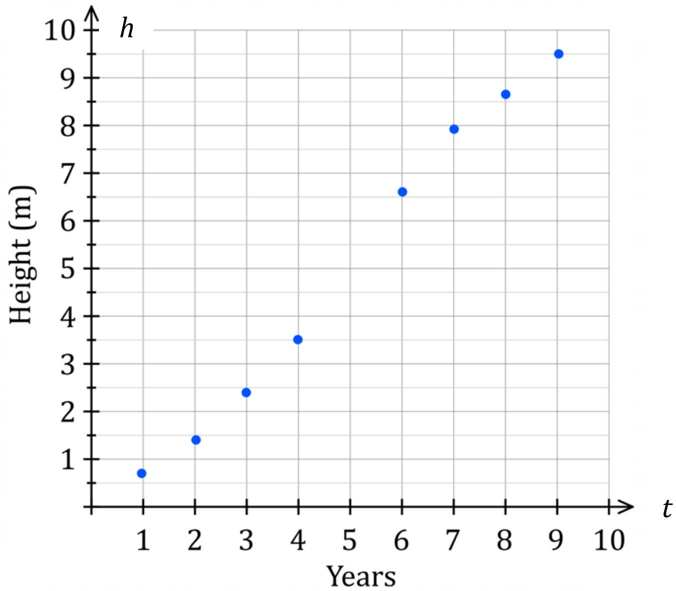
Question 29 (3 marks) **Marks**

Hayden is an agricultural scientist studying the growth of a particular tree over several years. The data he recorded is shown in the table below.

Years since planting, t	1	2	3	4	5	6	7	8	9
Height of tree, h metres	0.7	1.4	2.4	3.5	$\boldsymbol{H}$	6.6	7.9	8.7	9.5

$\boldsymbol{H} > 0$

A scatterplot of the data is shown below.



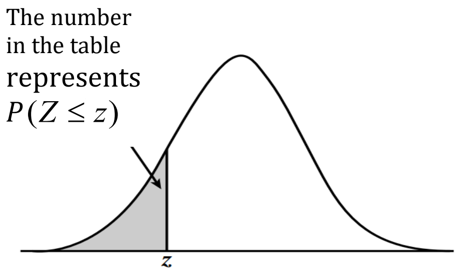
(a) What is Pearson's correlation coefficient? Answer correct to 4 decimal places. **1**

(b) Find the equation of the least-squares line of best fit in terms of years (t) and height (h). Answer correct to 2 decimal places. **1**

(c) Hayden did not record the tree's height after five years. Predict the height, $\boldsymbol{H}$, after five years, correct to one decimal place. **1**

Question 30 (2 marks) **Marks**

A section of the standard normal table is shown below:



Z	.00	.01	.02	.03	.04	.05
−2.4	.0082	.0080	.0078	.0075	.0073	.0071
−2.3	.0107	.0104	.0102	.0099	.0096	.0094
−2.2	.0139	.0136	.0132	.0129	.0125	.0122
−2.1	.0179	.0174	.0170	.0166	.0162	.0158
−2.0	.0228	.0222	.0217	.0212	.0207	.0202
−1.9	.0287	.0281	.0274	.0268	.0262	.0256
−1.8	.0359	.0351	.0344	.0336	.0329	.0322
−1.7	.0446	.0436	.0427	.0418	.0409	.0401
−1.6	.0548	.0537	.0526	.0516	.0505	.0495

Find $P\left(-2.3 \le z \le 1.72\right)$ **2**

Question 31 (6 marks) **Marks**

The height, $h(t)$, metres of the tide above the **mean sea level** on 1st April is given by the equation:

$$h(t) = 4\sin\left(\frac{\pi}{8}t\right)$$

where t is the number of hours after midnight.

(a) Draw a graph of $y = h(t)$ for $0 \le t \le 24$. **2**

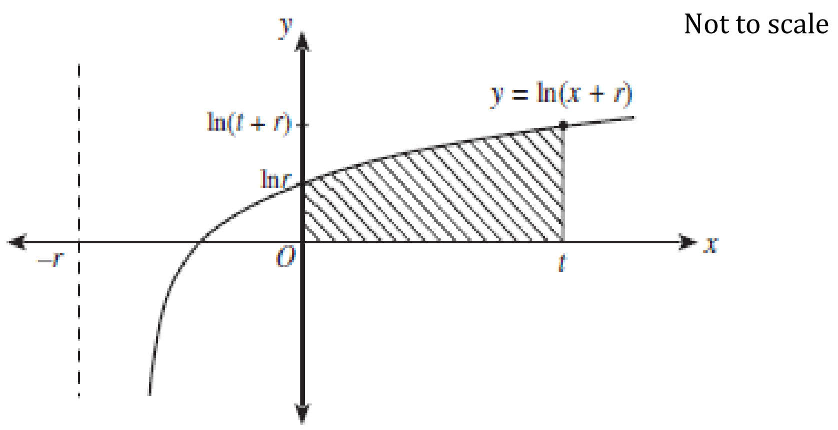
(b) When was high tide? **1**

(c) What was the height of the high tide? **1**

(d) What was the height of the tide at 10 am? Answer correct to one decimal place. **2**

Question 32 (3 marks) **Marks**

Consider the graph of $y = \ln(x + r)$ shown below. Let $r > 1$.



The region bound by the curve $y = \ln(x + r)$, the coordinate axes and the line $x = t$ is used as a model for a garden, where $t > 0$.

(a) Show that $x = e^y - r$ **1**

(b) Show that the area of this region, A, in terms of r and t is **2**

$$A = (t + r)\ln(t + r) - r\ln r - t$$

Question 33 (3 marks) **Marks**

The average life cycle of an insect is one month. A viable nest of this insect has between 100 000 to 500 000 insects. The population, *P,* of a nest of this insect grows exponentially so that: **3**

$$\frac{dP}{dt} = 1200e^{0\cdot 3t}$$

A nest of these insects had a population of 5 000 after one month.
Determine how long it will take the nest to reach the viable stage (i.e. when the population has reached 100 000). Answer correct to the nearest month.

Question 34 (8 marks) **Marks**

The acceleration on a particle, P_1, in ms^{-2} is $\frac{d^2x}{dt^2} = e^{-t} + e^{-2t}$, after t seconds. Initially, the particle is $\frac{3}{4}$ m to the right of the origin, travelling at velocity $\frac{dx}{dt} = -\frac{3}{2}$ m/s.

(a) Show that the displacement of the particle is given by **2**

$$\boldsymbol{x} = e^{-t} + \frac{1}{4}e^{-2t} - \frac{1}{2}\,.$$

(b) Find the limiting displacement of P_1 **and** hence state the limiting distance that P_1 travels. **2**

Question 34 (continued) **Marks**

(c) Another particle, P_2, moves simultaneously with the first particle, P_1. 4
The acceleration P_2 experiences is given by

$$\frac{d^2x}{dt^2} = -e^{-t} - e^{-2t}$$

This particle is also $\frac{3}{4}$ m to the right of the origin, travelling at velocity $\frac{dx}{dt} = -\frac{3}{2}$ m/s

when $t = \ln 3$ seconds.

Determine the exact time when P_1 and P_2 are travelling at the same velocity. Do NOT simplify your answer.

End of paper